1. Introduction

1.1 Background and Context

Today, most companies tend to automate different stages of the recruitment process. Many recruitment platforms generate an overwhelming amount of job offers daily that is rather difficult to manually analyze and compare to select one suitable position. Moreover, nowadays companies employ Applicant Tracking Systems (ATS) and automatic evaluation techniques to assess candidates' applications prior to their consideration by HR managers [1], [2].

However, the rapid progress of artificial intelligence and natural language processing technologies makes it possible to create novel solutions that will make the recruitment-related processes faster and more efficient. Large Language Models are now widely applied in various text analysis tasks, recommendation systems, semantic search engines, and document generation systems [3], [2]. It means that large amounts of text can be analyzed to establish logical relationships between user profiles and jobs descriptions.

Job recommendations are now an area where significant progress was achieved in terms of both research and engineering. Most job platforms utilize keyword-based filters but do not consider semantic similarities of candidate skill sets and job requirements [1]. As a result, some users might receive irrelevant job offers or have problems with passing the pre-selection stage of the application process because their resumes do not match the criteria specified by automated systems.

This thesis focuses on the development of an intelligent job recommendation and CV personalization system based on artificial intelligence methods and Large Language Models. The proposed solution aims to combine semantic job ranking with automated resume adaptation in order to improve the relevance of recruitment recommendations and increase compatibility with modern recruitment systems.

1.2 Motivation and Problem Statement

An increasing number of platforms for finding vacancies makes the job searching process even more difficult. As a rule, candidates have to manually evaluate numerous job offers, and only a fraction of all these positions suit their profile and professional goals. Also, many resumes get automatically filtered out since they either do not have appropriate keywords or are not properly customized.

There are several drawbacks of existing recommendation systems. First, keyword-driven approaches may fail to comprehend the contextual meaning of candidates' technical skills, work

experience descriptions, and job requirements [1]. Second, modern resume optimization tools are often unable to produce highly tailored content based on the semantic structure of the particular job offer.

The latest developments in the area of artificial intelligence, particularly the emergence of Large Language Models, may help address this issue. Namely, the ability of LLMs to interpret natural language, extract key facts from the document, and generate personalized texts allows leveraging this functionality to personalize resumes and rank job vacancies intelligently.

The problem addressed in this thesis is the lack of systems that combine intelligent job recommendation with automated CV adaptation using modern AI technologies. The proposed solution aims to reduce the time required for job searching and improve the quality of candidate-job matching through semantic analysis and automated text generation.

1.3 Aim and Objectives

The primary goal of this thesis is to create a smart recruitment support system that can make recommendations about job vacancies and generate customized CVs based on Large Language Models and natural language processing algorithms.

The goals of this thesis are stated below:

- to analyze existing approaches to job recommendation systems and resume evaluation
- to build a structure of the system for intelligent recruitment assistance
- to implement semantic ranking of job offers based on candidates' CVs and job descriptions
- to develop a CV personalization module using Large Language Models
- to report Precision@10 and nDCG@10 metrics for semantic ranking versus a keyword baseline on a fixed test set of job-CV pairs
- to explore the potential of AI-based methods in the current recruitment process

1.4 Scope of the Study

This thesis is strictly concerned with the creation of a software prototype solution based on intelligent job recommendations and personalized resumes. The main emphasis of this thesis lies in backend logic, semantic text processing, and AI-powered recommendation engines.

The proposed solution will be responsible for parsing user inputs regarding their skills, preferred programming languages, experience level, and desired jobs. The process of obtaining external job offers relies on publicly accessible API calls instead of web scraping techniques.

This thesis excludes any form of automation of the hiring process, like job application submissions or interactions with other recruitment websites via browser automation scripts. Moreover, the solution does not aim to automate the role of human recruiters or ensure the success of job applications.

The core of this research lies within semantic matching, intelligent ranking, and resume generation.

1.5 Thesis Structure

This thesis consists of five chapters.

Chapter 1 provides an overview of the background of the study. In this chapter, the rationale, problem statement, research objectives, and the scope of work are explained.

Chapter 2 provides a review of the theoretical background associated with recommendation systems, natural language processing techniques, Large Language Models, and the latest recruitment systems technologies. The existing works in the field and literature are also discussed.

Chapter 3 gives an overview of the proposed system design and its architecture. In this chapter, the technologies and software components used, and the functioning of recommendation and resume adaptation mechanisms are explained.

Chapter 4 analyzes the results of experiments conducted on the designed recommendation system. The functionality of the system for ranking jobs and adapting resumes is tested for effectiveness.

Chapter 5 contains the summary of the thesis, along with the limitations of the project and recommendations for future research.

References

1. R. V. K. Bevara, P. Kumar, P. S. Reddy, et al., *"Resume2Vec: Transforming Applicant Tracking Systems with Intelligent Resume Embeddings for Precise Candidate Matching,"* Electronics, vol. 14, no. 4, p. 794, 2025. DOI: 10.3390/electronics14040794

2. S. Vaishampayan, H. Leary, Y. Alebachew, et al., *"Human and LLM-Based Resume Matching: An Observational Study,"* Findings of the Association for Computational Linguistics: NAACL 2025. DOI: 10.18653/v1/2025.findings-naacl.270

3. X. Yu, R. Xu, C. Xue, J. Zhang, X. Ma, and Z. Yu, *"ConFit v2: Improving Resume-Job Matching using Hypothetical Resume Embedding and Runner-Up Hard-Negative Mining,"* Findings of the Association for Computational Linguistics: ACL 2025. DOI: 10.18653/v1/2025.findings-acl.661